



p-block elements (Boron and Carbon Family)

Section (A), (B) & (C) : General facts about elements, Based on Periodic trends & Based on Chemical Bonding

Introduction :

Group 13 to 18 of the periodic table of elements constitute the p-block. The p-block contains metals, metalloids as well as non-metals.

Configuration	ns^2np^1	ns^2np^2	ns^2np^3	ns^2np^4	ns^2np^5	ns^2np^6
						He
	B	C	N	O	F	Ne
	Al	Si	P	S	Cl	Ar
	Ga	Ge	As	Se	Br	Kr
	In	Su	Sb	Te	I	Xe
	Tl	Pb	Bi	Po	At	Rn
Group Oxidation State	+3	+4	+5	+6	+7	+8
Other Oxidation State	+1	+2, -4	+3, -3	+4, +2, -2	+5, +3, +1, -1	+6, +4, +2

The p-block elements have general valence shell electronic configuration $ns^2 np^{1-6}$.

The first member of each group from 13–17 of the p-block elements differ in many respects from the other members of their respective groups because of small size, high electronegativity and absence of d-orbitals.

The first member of a group also has greater ability to form $p\pi-p\pi$ multiple bonds to itself (e.g. $C=C$, $C\equiv C$, $N\equiv N$) and to element of second row (e.g. $C=O$, $C=N$, $C\equiv N$, $N=O$) compared to the other members of the same group.

The highest oxidation state of p-block element is equal to the group number –10. Moving down the group, the oxidation state two less than the highest group oxidation state and becomes more stable in groups 13 to 16 due to inert pair effect (reluctance of s-subshell electrons to participate in chemical bonding)

- Ge liquid expands when it forms the solid. This property is unique to Ga, Ge and Bi.

Catenation :

Carbon atoms have the tendency to link with one another through covalent bonds to form chains and rings. This property is called catenation. This is because C–C bonds are very strong. Down the group the size increases tendency to show catenation decreases. This can be clearly seen from bond enthalpies values. The order of catenation is $C \gg Si > Ge \approx Sn$. Lead does not show catenation. Due to the property of catenation and $p\pi-p\pi$ bonds formation, carbon is able to show allotropic forms.

Bond	Bond enthalpy (kJ mol^{-1})
C—C	348
Si—Si	297
Ge—Ge	260
Sn—Sn	240

ANOMALOUS BEHAVIOUR OF CARBON :

Like first member of other groups, carbon also differs from rest of the members of its group. It is due to its smaller size, higher electronegativity, higher ionisation enthalpy and unavailability of d orbitals.

Carbon accommodate only four pairs of electrons around it and thus this would limit the maximum covalence to four whereas other members can expand their covalence due to the presence of d orbitals, Carbon also has unique ability to form $p\pi-p\pi$ multiple bonds with itself and with other atoms of small size and high electronegativity. Few examples of multiple bonding are $C=C$, $C\equiv C$, $C=O$, $C=S$ and $C\equiv N$. Heavier elements do not form $p\pi-p\pi$ bonds because their atomic orbital are too large and diffuse to have effective overlapping.



IMPORTANT TRENDS AND ANOMALOUS PROPERTIES OF BORON

The tri-chlorides, bromides and iodides of all these elements being covalent in nature are hydrolysed in water. Species like tetrahedral $[M(OH)_4]^-$ and octahedral $[M(H_2O)_6]^{3+}$, except in boron, exist in aqueous medium. It is due to the absence of d orbitals that the maximum covalence of boron is 4. Since the d-orbitals are available with Al and other elements, the maximum covalence can be expected beyond 4.

GROUP 13 : BORON FAMILY & GROUP 14 : CARBON FAMILY

Occurrence :

Element	Abundance	Source	Element	Abundance	Source
B (Boron)	9 ppm (Rare element)	- Borax : $Na_2[B_4O_5(OH)_4] \cdot 8H_2O$ - Colemanite : $Ca_2B_6O_{11} \cdot 5H_2O$ - Kernite : $Na_2[B_4O_5(OH)_4] \cdot 2H_2O$ - Boric acid : H_3BO_3	C (Carbon)	180 ppm	- Coal - Natural gas - oil (hydrocarbon) - Natural Graphite - Natural diamond - Carbonates: Calcite($CaCO_3$) - Magnesite ($MgCO_3$) - Dolomite($MgCO_3 \cdot CaCO_3$)
Al (Aluminium)	83000 ppm (Most abundant metal, 3 rd most Abundant element)	- Bauxite : $Al_2O_3 \cdot H_2O - Al_2O_3 \cdot 3H_2O$ - Aluminosilicate rocks (feldspars, mica) - Cryolite : Na_3AlF_6	Si (Silicon)	272000 ppm	- Silica (sand & quartz) SiO_2 - Silicate minerals
Ga (Gallium)	19 ppm	• Ores of Al, Zn, Ge	Ge (Germanium)	1.5 ppm	• Silver and Zinc ores, coal
In (Indium)	0.24 ppm	• ZnS & PbS ores	Sn (Tin)	2.1 ppm	• Cassiterite (SnO_2)
Tl (Thallium)	0.5 ppm	• ZnS & PbS ores	Pb (Lead)	13 ppm	• Galena (PbS)

Allotropy :

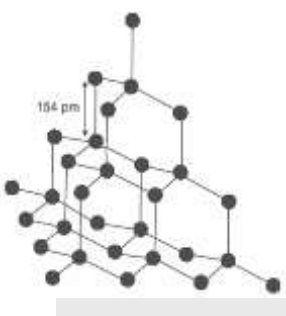
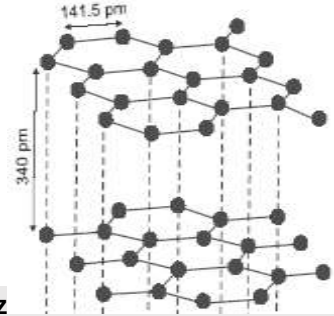
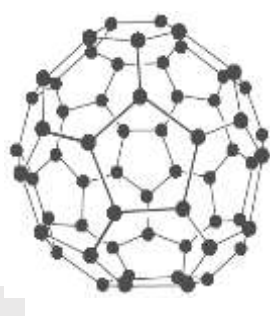
Elements	Allotropes	Elements	Allotropes
B	5 crystalline forms : β -rhombohedral	C	Crystalline Diamond, hexagonal diamond
	2 amorphous forms		α -graphite, β -graphite fullerenes
Al	No allotrope	Si	Amorphous - brown powder Crystalline - greyish metallic
Ga	No allotrope	Ge	α -Germanium
			β -Germanium
			γ -Germanium
In	No allotrope	Sn	α -tin (grey tin)
			β -tin (white tin)
Tl	α -Thallium	α -Sn(diamond structure) $\xrightleftharpoons{13.2^\circ C}$ β -Sn(metallic)	
	β -Thallium	Pb	α -lead (Pb-I)
			β -lead (Pb-II)



Allotropy of carbon family

○ Silicon is a very high melting-point solid with the same structure as diamond. The non-existence of an allotrope with the graphite structure clearly shows the inability of silicon atoms to multiple bond with themselves.

Allotropes of Carbon:

	Diamond	Graphite	Fullerene
Structure			
Hybridisation	sp^3	sp^2	sp^2
Density (g/cm^3)	3.51	2.22	1.65
ΔH_f (KJ/mol)	1.9	0	38.1
Bond length	154 pm	141.5 pm	143.5 pm & 138.3 pm
	- Crystalline lattice. 3-D network: each C-atom is linked to four other C-atoms in tetrahedral manner. - One of the hardest – next to boron nitride (only at certain conditions) - Uses: sharpening hard tools, cutter tools; as a gem.	- Layered structure - Interlayer force-Vanderwaal's forces - Each carbon atom is linked to three other carbon atoms, fourth electron forms a π bond. - Good conductor along the sheet and semi-conductor perpendicular to the sheet. - Inter layer distance 340 pm so cleavage between layers is easy. Soft and slippery– lubricant at high temperature. - Natural graphite is found as a mixture with mica, quartz & silicates. $3C + SiO_2 \xrightarrow{\Delta} SiC + 2CO \xrightarrow{2500^\circ C} C(\text{graphite}) + Si \text{ gas}$ - Thermodynamically most stable among allotropes. $Graphite \xrightarrow[50000-60000 \text{ atm}]{1600^\circ C} \text{synthetic diamond}$	- Cage like molecules. C_{60}–Soccer ball shape–Buckminsterfullerene 20- six membered rings and 12- five membered rings. - Six membered ring is fused with six or five membered ring - Five membered ring is fused only with six membered ring. - Heating of graphite in an electrical arc in the presence of inert gases such as helium or argon can result into fullerene

Solved Examples

Ex-1. Thermodynamically graphite is more stable than diamond but diamond does not transform into graphite on its own. Why?

Sol. This conversion is not favoured by kinetic factors (the activation energy for this is very high).



Atomic & Physical Properties

Boron is non-metallic in nature. It is extremely hard and black coloured solid. It exists in many allotropic forms. Due to very strong crystalline lattice, boron has unusually high melting point. Rest of the member are soft metals with low melting point and high electrical conductivity. Gallium with low melting point (303 K), could exist in liquid state during summer. Its high boiling point (2676 K) makes it a useful material for measuring high temperatures. Aluminium is a good conductor of heat and electricity. It is malleable and ductile. Density of the elements increases down the group from boron to thallium.

Boron family:

Property			B	Al	Ga	In	Tl
Character			Metalloid	Metallic	Metallic	Metallic	Metallic
Atomic Number			5	13	31	49	81
Atomic Mass/g mol ⁻¹			10.81	26.98	69.72	114.82	204.38
Electronic configuration General electronic configuration=(ns ² np ¹)			[He] 2s ² 2p ¹	[Ne] 3s ² 3p ¹	[Ar] 3d ¹⁰ 4s ² 4p ¹	[Kr] 4d ¹⁰ 5s ² 5p ¹	[Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ¹
Covalent Radius / pm (B < Ga < Al < In < Tl)			85	143	135	167	170
			In Ga, poor shielding of 10 d-electrons				
Ionic Radius X ⁻ / pm (B < Al < Ga < In < Tl)			27	53.5	62	80	88.5
Ionization enthalpy (kJ mol ⁻¹) (B > Al < Ga > In < Tl)	$\Delta_i H_1$	I	801	577	579	558	589
	$\Delta_i H_2$	II	2427	1816	1979	1820	1971
	$\Delta_i H_3$	III	3659	2744	2962	2704	2877
			Poor shielding of d-orbital and f-orbital in Ga & Tl respectively				
Electronegativity (B > Al < Ga < In < Tl)			2.0	1.5	1.6	1.7	1.8
			Marginal increase after Al				
Melting point / K			2453	933	303	430	576
Boiling point / K (B > Al > Ga > In > Tl)			3923	2740	2676	2353	1730
Density/[g cm ⁻³ (at 293 K)] (B < Al < Ga < In < Tl)			2.35	2.70	5.90	7.31	11.85
Oxidation State			+3	+3	+3, +1	+3, +1	+3, +1
			+1 oxidation state arises due to inert pair effect.				
			Stability of Oxidation state: +1: Ga < In < Tl +3 : Al > Ga > In > Tl				

Carbon family :

Property	C	Si	Ge	Sn	Pb
Character	Non Metallic	Non Metallic	Metalloid	Metallic	Metallic
Atomic Number	6	14	32	50	82
Atomic Mass/g mol ⁻¹	12.01	28.09	72.60	118.71	207.2
Electronic configuration General electronic configuration=(ns ² np ²)	[He] 2s ² 2p ²	[Ne] 3s ² 3p ²	[Ar] 3d ¹⁰ 4s ² 4p ²	[Kr] 4d ¹⁰ 5s ² 5p ²	[Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ²



Covalent Radius / pm (C < Si < Ge < Sn < Pb)			77	118	122	140	146
Ionic Radius M ⁴⁺ / pm (Si < Ge < Sn < Pb)			–	40	53	69	78
Ionization enthalpy (kJ mol ⁻¹) (C > Si > Ge > Sn < Pb)	$\Delta_i H_1$	I	801	577	579	558	589
	$\Delta_i H_2$	II	2427	1816	1979	1820	1971
	$\Delta_i H_3$	III	3659	2744	2962	2704	2877
Poor shielding f-orbital							
Electronegativity (C > Si ≈ Ge ≈ Sn ≈ Pb)			2.5	1.8	1.8	1.8	1.9
Melting point / K			4373	1693	1218	505	600
Boiling point / K			–	3550	3123	2896	2024
Density/[g cm ⁻³ (at 293 K)] (C (diamond) > Si < Ge < Sn < Pb)			3.51	2.34	5.32	7.26	11.34
Oxidation State			+4	+4	+4, +2	+4, +2	+4, +2
			+2 oxidation state arises due to inert pair effect.				
			Stability of Oxidation state: +2: Ge < Sn < Pb +4: Ge > Sn > Pb				

The Elements

Preparation of elements :

BORON :

Source	Process	Comments
From Borax (Na ⁺) ₂ B ₄ O ₇ ·10H ₂ O	- Na₂B₄O₇ + 2HCl + 5H₂O → 4H₃BO₃ + 2NaCl 2H₃BO₃ $\xrightarrow{\Delta}$ B₂O₃ + 3H₂O - B₂O₃ + 3Mg/Na $\xrightarrow{\text{High temp.}}$ 2B + 3MgO/Na₂O	- low purity (95-98%) boron (black) - The product thus obtained is boiled with HCl and filtered when Na₂O or MgO dissolves leaving behind elemental boron. It is thoroughly washed to remove HCl and then dried finally.
From BX ₃	2BX₃ + 3H₂ $\xrightarrow[\text{or Tantalum}]{\text{red hot W}}$ 2B(crystalline) + 6HX (X = Cl or Br) 99.9% pure 2BCl₃ + 3Zn → 3ZnCl₂ + 2B 2BI₃ $\xrightarrow[\text{or Tantalum}]{\text{red hot W}}$ 2B↑ + 3I₂↑ (Van Arkel method)	Problem in obtaining high purity boron: - High melting point (2180°C) - Liquid gets corroded
From Diborane (B ₂ H ₆)	B₂H₆ $\xrightarrow{\Delta}$ 2B (crystalline) + 3H₂↑	Thermal decomposition of diborane.
From potassium fluoroborate (KBF ₄)	KBF₄ + 3K $\xrightarrow{\Delta}$ 4KF + B	- By heating it with potassium metal. - It is then treated with dilute HCl to remove KF and B is then washed and dried.

**ALUMINIUM :**

Method	Process	Comments
Electrolytic reduction (Hall-Heroult process)	The electrolytic reactions are: Cathode : $\text{Al}^{3+}(\text{melt}) + 3\text{e}^- \longrightarrow \text{Al}(\text{l})$ Anode : $\text{C}(\text{s}) + \text{O}^{2-}(\text{melt}) \longrightarrow \text{CO}(\text{g}) + 2\text{e}^-$ $\text{C}(\text{s}) + 2\text{O}^{2-}(\text{melt}) \longrightarrow \text{CO}_2(\text{g}) + 4\text{e}^-$ The overall reaction may be taken as: $2\text{Al}_2\text{O}_3 + 3\text{C} \longrightarrow 4\text{Al} + 3\text{CO}_2$ Several other extraction processes will be studied in detail in Metallurgy.	- The purified Al_2O_3 is mixed with Na_3AlF_6 (cryolite) or CaF_2 (fluorspar) which lowers the melting point of the mixture and brings conductivity. The fused matrix is electrolysed. Steel cathode and graphite anode are used. The graphite anode is useful here for reduction to the metal. - The electrolysis of the molten mass is carried out in an electrolytic cell using carbon electrodes. The oxygen liberated at anode reacts with the carbon of anode producing CO and CO_2. This way for each kg of aluminium produced, about 0.5 kg of carbon anode is burnt away.

CARBON :

Preparation of	Process
Carbon black (soot)	By incomplete combustion of hydrocarbon.
Graphite	$3\text{C} + \text{SiO}_2 \xrightarrow{\Delta} \text{SiC} + 2\text{CO} \xrightarrow{2500^\circ\text{C}} \text{C}(\text{graphite}) + \text{Si gas}$
Diamond	- Natural diamond can be extracted from mines. - Synthetic diamond $\text{Graphite} \xrightarrow[50000-60000 \text{ atm}]{1600^\circ\text{C}} \text{synthetic diamond.}$

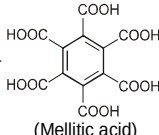
SILICON :

From	Process	Comments
SiO_2	$\text{SiO}_2(\text{excess}) + 2\text{C} \longrightarrow \text{Si}(\text{pure}) + 2\text{CO}$ $\text{Si}(\text{pure}) + 2\text{Cl}_2 \longrightarrow \text{SiCl}_4$ $\text{SiCl}_4 + 2\text{Mg} \longrightarrow \text{Si}(\text{highly pure}) + \text{MgCl}_2$	Reducing SiO_2 with high purity coke in an electric furnace
$\text{Na}_2[\text{SiF}_6]$	Sodium Reduction of $\text{Na}_2[\text{SiF}_6]$ $\text{Na}_2[\text{SiF}_6] + 4\text{Na} \longrightarrow \text{Si}(\text{ultra pure}) + 6\text{NaF}$	Zone refining is used to get ultra pure silicon from highly pure silicon, which can be used in semi-conductor industry.



Section (D) : Properties of Elements

Chemical Properties :

Reaction with	Boron Family			Carbon Family				
O ₂	4E + 3O ₂ $\xrightarrow{\Delta}$ 2E ₂ O ₃			E + O ₂ $\xrightarrow{\Delta}$ EO ₂ (dioxide) E + $\frac{1}{2}$ O ₂ $\xrightarrow{\Delta}$ EO (monoxide)				
	B ₂ O ₃ Al ₂ O ₃ Ga ₂ O ₃ In ₂ O ₃ Tl ₂ O ₃	– – – – –	acidic amphoteric amphoteric basic basic	CO ₂ SiO ₂ GeO ₂ SnO ₂ PbO ₂	– – – – –	acidic acidic acidic amphoteric amphoteric	CO SiO GeO SnO PbO	– – – – –
N ₂	2E + N ₂ $\xrightarrow{\Delta}$ 2EN EN + H ₂ O $\xrightarrow{\Delta}$ E(OH) ₃ + NH ₃ ∴ E = B or Al			2C(s)+O ₂ (g)+4N ₂ (g) $\xrightarrow{1273\text{K}}$ 2CO(g) + 4N ₂ (g) (Produce gas)				
X ₂ (Halogen)	2E + 3X ₂ $\longrightarrow$ 2EX ₃ ; (X=F, Cl, Br, I)			E + 2X ₂ $\longrightarrow$ EX ₄ ; (X = F, Cl, Br, I)				
	BX ₃ AlF ₃ GaF ₃ InF ₃ TlI ₃	– – – – –	covalent (<i>BI₃ cannot be formed directly</i>) ionic ionic ionic exist as Tl ⁺ I ₃ [–] & TlI also forms	- All members form MX₄ ; Ge & Pb form MX₂ - PbI₄ does not exist - Stability of EX₄ decreases down the group - Stability of EX₂ increases down the group - Stability : GeX₄ > GeX₂ ; PbX₂ > PbX₄				
H ₂ O	B – does not react with water Al $\xrightarrow[+H_2O]{25^\circ C}$ Al(OH) ₃ $\xrightarrow[+H_2O]{>480^\circ C}$ Al ₂ O ₃ Ga $\xrightarrow{\quad}$ Not attacked by cold & hot water unless oxygen is present. In $\xrightarrow{\quad}$ Not attacked by cold & hot water unless oxygen is present. Tl – oxidises in moist air & decomposes steam at red heat.			C(s) + H ₂ O (steam) $\xrightarrow[\text{heat}]{\text{red}}$ CO(g) + H ₂ (g) (water gas) E(s) + H ₂ O(steam) $\xrightarrow[\text{heat}]{\text{red}}$ EO ₂ +H ₂ ; (Si,Ge,Sn) - Pb is unaffected by water, probably because of a protective oxide layer. - C, Si, Ge – not attacked by cold water.				
Acids	B $\xrightarrow[+HNO_3]{+H_2SO_4 \text{ (Hot \& Conc.)}}$ H ₃ BO ₃ + SO ₂ $\xrightarrow[+HNO_3]{\quad}$ H ₃ BO ₃ + NO ₂ (Boron reacts with only oxidising acids) Al $\xrightarrow{+HCl}$ AlCl ₃ + H ₂ $\xrightarrow[+HNO_3]{\quad}$ Do not react because it forms passive oxide layer. Ga, In, Tl can also react with dilute mineral acids.			C $\xrightarrow[+HCl]{\quad}$ No reaction Si $\xrightarrow[+HCl]{\quad}$ No reaction Ge $\xrightarrow[+HCl]{\quad}$ No reaction Sn dissolves in HCl (dil. & conc) but Pb in only dil.HCl. Pb $\xrightarrow[+H_2SO_4]{HCl \text{ (conc.)}}$ PbCl ₂ + H ₂ (coating) (does not dissolve in) $\xrightarrow{\quad}$ PbSO ₄ + H ₂ (coating) C (graphite) $\xrightarrow[+HF/HNO_3]{+HNO_3 \text{ (Hot \& conc.)}}$  (Mellitic acid) $\xrightarrow[+HF/HNO_3]{\quad}$ Graphite oxide (mixture) Si + 6HF $\rightarrow$ H ₂ SiF ₆ + 2H ₂ E + dil. HNO ₃ $\rightarrow$ E(NO ₃) ₂ (Sn,Pb)				



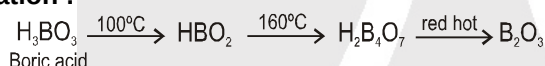
NaOH	$2B + 2NaOH + 2H_2O \longrightarrow 2NaBO_2 + 3H_2$ $2B + 6NaOH \xrightarrow{\text{fused}} 2Na_3BO_3 + 3H_2$ $Al \xrightarrow[NaOH]{NaOH + H_2O} NaAlO_2 \cdot 2H_2O + H_2$ or $Na^+[Al(OH)_4]^-$ $\xrightarrow{NaOH} Na_3AlO_3 + H_2$ <i>(Al & Ga readily dissolves in alkalis).</i> <i>In & Tl do not react with alkali.</i>	C – unaffected by alkalis $Si + 2NaOH + H_2O \longrightarrow Na_2SiO_3 + 2H_2$ $E + NaOH \xrightarrow{\Delta} Na_2EO_3 + H_2$ or $Na_2[E(OH)_6]$ <i>(Sn, Pb)</i>
	Metal $3Mg + 2B \rightarrow Mg_3B_2$ $3Ca + 2B \rightarrow Ca_3B_2$	$2Mg + Si \xrightarrow{\Delta} Mg_2Si$ (magnesium silicide)
Reducing Property	$B \xrightarrow[\Delta]{SiO_2} B_2O_3 + Si$ $\xrightarrow[\Delta]{CO_2} B_2O_3 + C$ $Al \xrightarrow[\Delta]{MnO_2} Al_2O_3 + Mn$ $\xrightarrow[\Delta]{Cr_2O_3} Al_2O_3 + Cr$	

Section (E) : Oxides, Hydroxides, Oxyacids, Borax

Oxides, oxy acids and hydroxide

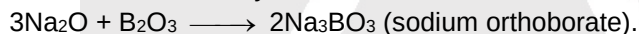
Boron trioxide (B₂O₃) :

Preparation :

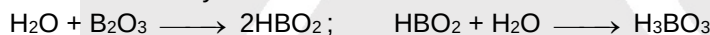


Properties :

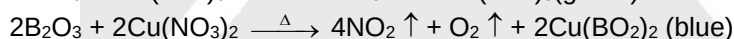
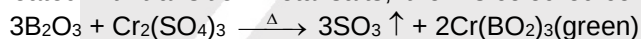
It is an acidic oxide and is an anhydride of boric acid and it reacts with alkalis or bases to form borates.



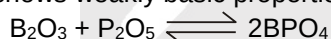
It reacts with water slowly to form orthoboric acid.



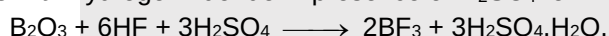
When heated with transition metal salts, it forms coloured compounds.



It also shows weakly basic properties according to the following reaction.



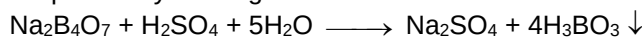
It reacts with hydrogen fluoride in presence of H₂SO₄ forming BF₃.



Boric acid (H₃BO₃) :

Preparation :

(i) It is precipitated by treating a concentrated solution of borax with sulphuric acid.

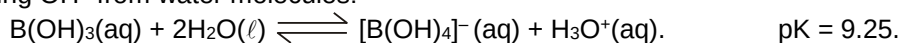


(ii) From Colemanite: Powdered colemanite is suspended in water and excess SO₂ is passed through it. On filtering and cooling the filtrate, white crystals of H₃BO₃ are obtained.



Properties:

It is a weak monobasic acid, soluble in water and in aqueous solution the boron atom completes its octet by accepting OH⁻ from water molecules:

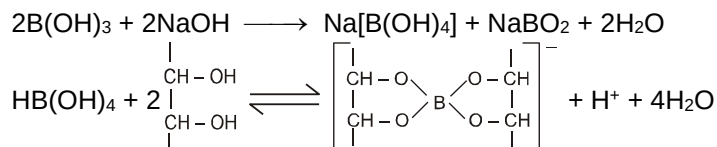


It, therefore, functions as a Lewis acid and not as a proton donor like most acids.

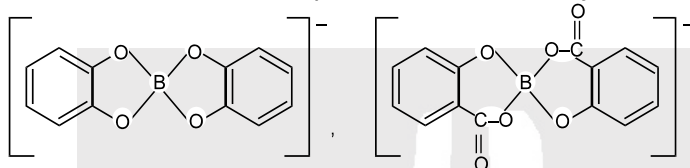


Since $B(OH)_3$ only partially reacts with water to form H_3O^+ and $[B(OH)_4]^-$, it behaves as a weak acid. Thus H_3BO_3 cannot be titrated satisfactorily with $NaOH$ as a sharp end point is not obtained. If certain organic polyhydroxy compounds such as glycerol, mannitol or sugars are added to the titration mixture, then $B(OH)_3$ behaves as a strong monobasic acid and it can be now titrated with $NaOH$ and the end point is detected using phenolphthalein as indicator ($pH = 8.3 - 10.0$).

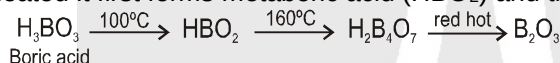
The added compound must be a cis-diol, to enhance the acid properties. The cis-diol forms very stable complex with the $[B(OH)_4]^-$, thus removing it from solution. The reaction is reversible and thus removal of one of the products shifts the equilibrium in the forward direction and thus all the $B(OH)_3$ reacts with $NaOH$; in effect it acts as a strong acid in the presence of the cis-diol.



Ethanol does not form similar complex but catechol, salicylic acids, mannitol form similar complexes.



When heated it first forms metaboric acid (HBO_2) and then boron trioxide.



Orthoboric acid is greasy to touch less soluble in cold water but more soluble in hot water. In the solid state, the $B(OH)_3$ units are hydrogen bonded together in to two dimensional sheets with almost hexagonal symmetry. The layered are quite a large distance apart ($3.18 \text{ \AA}$) and thus the crystal breaks quite easily into very fine particles.

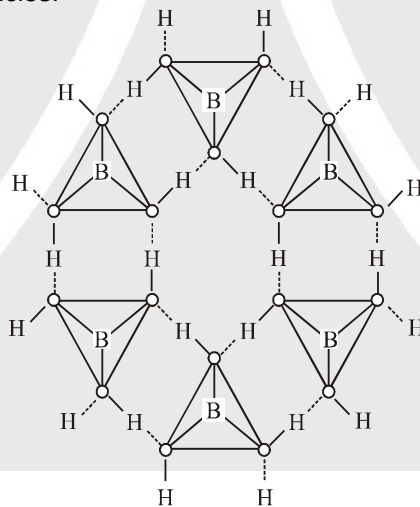
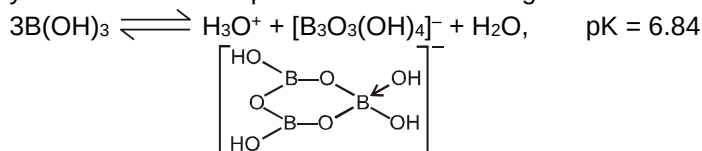
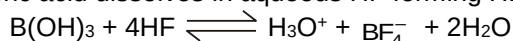


Figure : 1

- Polymeric metaborate species are formed at higher concentration, for example,

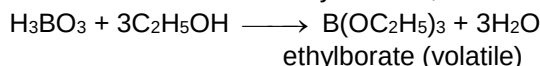


- Boric acid dissolves in aqueous HF forming HBF_4 (fluoroboric acid).



**Test for Borate radical :**

When boric acid is heated with ethyl alcohol, the evolved gas is burned forming a green edged flame.

**Uses :**

- (i) It is an antiseptic and its water solution is used as an eyewash.
- (ii) It is also used in glass, enamel and pottery industry.

Solved Examples

Ex-2. It has been observed that BF_3 does not hydrolyses completely whereas BCl_3 or BBr_3 get easily hydrolysed to form $\text{B}(\text{OH})_3$ and HX ? Explain.

Sol. The greater stability of B–F bond as compared to B–Cl and B–Br bonds is due to additional π -bonding in B–F bonds of BF_3 molecules. The B–Cl and B–Br bonds are relatively weak and are easily cleaved by water forming strong B–OH bonds instead of stable addition product $(\text{BF}_3 \cdot \text{OH}_2)$ formed by BF_3 .

Borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) :**Preparation :**

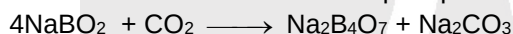
It is found in nature but can also be prepared by the following methods.

(i) From Colemanite.

When colemanite powder is heated with Na_2CO_3 solution, the following reaction occurs with the precipitation of CaCO_3 .

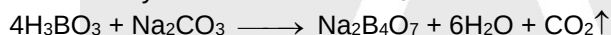


The filtrate is cooled when white crystals of borax are precipitated. The mother liquor on treatment with CO_2 converts NaBO_2 to $\text{Na}_2\text{B}_4\text{O}_7$ which precipitates out on crystallization.



(ii) From orthoboric acid.

Borax is obtained by the action of Na_2CO_3 on orthoboric acid.

**Properties :**

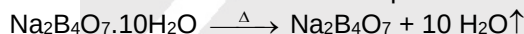
(i) Borax is a white powder, less soluble in cold water, more soluble in hot water.

(ii) Its aqueous solution is alkaline because of its hydrolysis to weak acid H_3BO_3 and strong alkali NaOH .

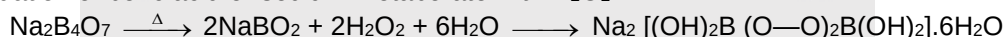


(iii) Action of heat.

When borax powder is heated, it first swells due to loss of water in the form of steam but at 740°C it becomes converted into colourless transparent borax bead.

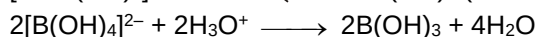
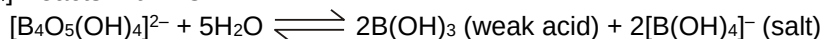


(iv) Oxidation of boric acid or sodium metaborate with H_2O_2 .



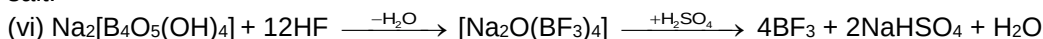
Sodium per oxoborate is used as a brightener in washing powder. In very hot water (over 80°C) the peroxide linkages $-\text{O}-\text{O}-$ break down to give H_2O_2 .

(v) It is a useful primary standard for titrating against acids. One mole of it reacts with two moles of acid. This is because when borax is dissolved in water both $\text{B}(\text{OH})_3$ and $[\text{B}(\text{OH})_4]^-$ are formed, but only the $[\text{B}(\text{OH})_4]^-$ reacts with HCl .



On cooling, the white flakes of boric acid are obtained

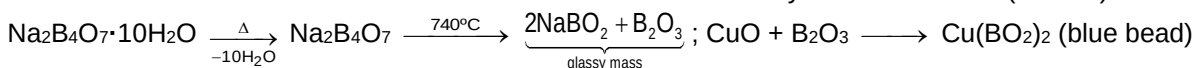
Borax is also used as a buffer since its aqueous solution contain equal amounts of weak acid and its salt.



Correct formula of borax is $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$. It contains boron in both planar BO_3 and tetrahedral BO_4 units. It contains five B–O–B linkages.

**Borax-bead test :**

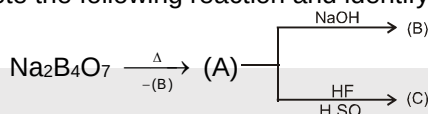
Borax reacts with certain metal salts such as, Ni^{2+} , Co^{2+} , Cr^{3+} , Cu^{2+} , Mn^{2+} etc. to form coloured metaborates. The colour of the metaborates can be used to identify the metallic ions (cations) in salts.

**Uses :**

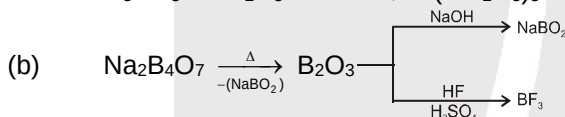
- (i) In borax bead test.
- (ii) In purifying gold.
- (iii) As flux during welding of metals.
- (iv) In production of glass.

Solved Examples

- Ex-.3** (a) $\text{Na}_2\text{B}_4\text{O}_7 + \text{concentrated H}_2\text{SO}_4 + \text{H}_2\text{O} \longrightarrow (\text{A}) \xrightarrow[\text{(ii) ignite}]{\text{(i) C}_2\text{H}_5\text{OH}} (\text{B})$
 (B) is identified by the characteristic colour of the flame. Identify (A) and (B).
 (b) Complete the following reaction and identify the products formed.



- Sol.** (a) $\text{Na}_2\text{B}_4\text{O}_7 + \text{concentrated H}_2\text{SO}_4 + 5\text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + 4\text{H}_3\text{BO}_3$
 $\text{H}_3\text{BO}_3 + 3\text{C}_2\text{H}_5\text{OH} \longrightarrow \text{B}(\text{OC}_2\text{H}_5)_3 \text{ -volatile (burn with green edged flame) } + 3\text{H}_2\text{O}$

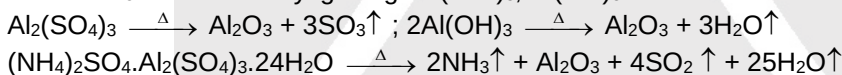
**Aluminium Oxide (Al_2O_3) :**

It is also called alumina. It occurs in nature in the form of bauxite and corundum. It is also found in the form of gems. Some important aluminium oxide gems are :

- (A) Oriental Topaz-yellow (Fe^{3+}),
- (B) Sapphire-blue ($\text{Fe}^{2+} / \text{Ti}^{4+}$),
- (C) Ruby-red (Cr^{3+}),
- (D) Oriental Emerald-green ($\text{Cr}^{3+} / \text{V}^{3+}$)

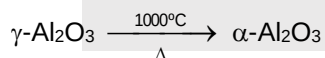
Preparation :

Pure Al_2O_3 is obtained by igniting $\text{Al}_2(\text{SO}_4)_3$, $\text{Al}(\text{OH})_3$ or ammonium alum.

**Properties :**

It is a white amorphous powder insoluble in water but soluble in acids (forming eg., AlCl_3) as well as alkalis (forming e.g., NaAlO_2). Thus amphoteric in nature. It is a polar covalent compound. Exists in two forms $\alpha\text{-Al}_2\text{O}_3$ or corundum and $\gamma\text{-Al}_2\text{O}_3$.

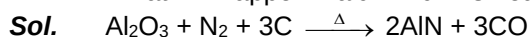
Addition of Cr_2O_3 or Fe_2O_3 makes alumina coloured.

**Uses :**

- (i) It is used for the extraction of aluminium.
- (ii) It is used for making artificial gems.
- (iii) It is used for the preparation of compounds of aluminium.
- (iv) $\alpha\text{-Al}_2\text{O}_3$ is used in making furnace linings. It is a refractory material.
- (v) It is used as a catalyst in organic reactions.
- (vi) Corundum is extremely hard and is used as 'Jewellers rouge' to polish glass.
- (vii) $\gamma\text{-Al}_2\text{O}_3$ dissolves in acids absorbs moisture and is used in chromatography.

Solved Examples

- Ex-4** What will happen if aluminium is heated with coke in an atmosphere of nitrogen ?

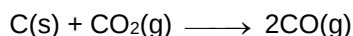




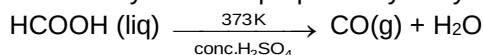
Carbonmonoxide (CO) :

Preparation :

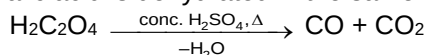
(i) It is formed together with CO₂, when carbon or carbonaceous matter is oxidized by air or oxygen. It is also produced when CO₂ is reduced by red-hot carbon; this reaction is of importance in metal extractions.



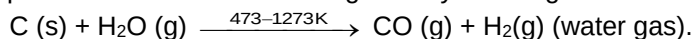
(ii) In the laboratory it can be prepared by dehydrating methanoic acid with concentrated sulphuric acid.



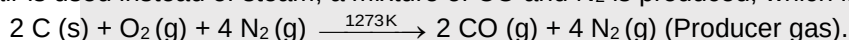
(iii) If oxalic acid is dehydrated in the same way, CO₂ is formed as well.



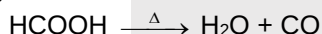
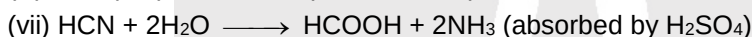
(iv) On commercial scale it is prepared by the passage of steam over hot coke. The mixture of CO and H₂ thus produced is known as water gas or synthesis gas.



When air is used instead of steam, a mixture of CO and N₂ is produced, which is called producer gas.



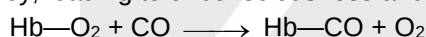
Water gas and producer gas are very important industrial fuels. Carbon monoxide in water gas or producer gas can undergo further combustion forming carbon dioxide with the liberation of heat.



(viii) Also obtained as by-product when carbon is used in reduction processes such as, of phosphite rock to give phosphorus.

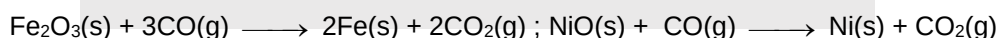
Properties :

(i) Carbon monoxide is a colourless, odourless gas which burns in air with a blue flame, forming CO₂. It is sparingly soluble in water and is a neutral oxide. CO is toxic, because it forms a complex with haemoglobin in the blood and this complex is more stable than oxy-haemoglobin. This prevents the haemoglobin in the red blood corpuscles from carrying oxygen round the body. This causes oxygen deficiency, leading to unconsciousness and then death.



Ordinary gas masks are no protection against the gas, since it is not readily adsorbed on active charcoal. In the presence of air, a mixture of manganese (IV) oxide and copper(II) oxide catalytically oxidizes it to CO₂, and this mixed catalyst is used in the breathing apparatus worn by rescue teams in mine disasters.

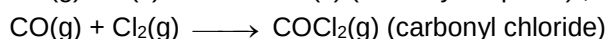
(ii) Carbon monoxide is a powerful reducing agent, being employed industrially in the extraction of iron and nickel.



(iii) It reacts with many transition metals, forming volatile carbonyls; the formation of nickel carbonyl followed by its decomposition is the basis of the Mond's process for obtaining very pure nickel.



(iv) In addition to reacting with oxygen, carbon monoxide combines with sulphur to give carbonyl sulphide and with chlorine in the presence of light to give carbonyl chloride (phosgene), used in the production of polyurethane foam plastics. Phosgene is an exceedingly poisonous gas.

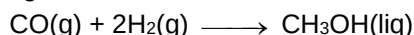


(v) Although carbon monoxide is not a true acid anhydride since it does not react with water to produce an acid, it reacts under pressure with fused sodium hydroxide to give sodium methanoate :





(vi) With hydrogen under pressure and in the presence of zinc oxide or chromium (III) oxide catalyst it reacts to give methanol; this reaction is of industrial importance.



(vii) CO is readily absorbed by an ammonical solution of copper (I) chloride to give $\text{CuCl} \cdot \text{CO} \cdot 2\text{H}_2\text{O}$. It reduces an ammonical solution of silver nitrate to silver (black) and, in the absence of other gaseous reducing agents, this serves as a test for the gas. It can be estimated by reaction with iodine pentoxide, the iodine which is produced quantitatively being titrated with standard sodium thiosulphate solution.

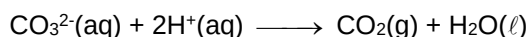


(viii) It reduces an aqueous PdCl_2 solution to metallic Pd.

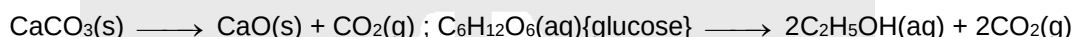
Carbon dioxide (CO_2) :

Preparation :

(i) In the laboratory it can be conveniently made by the action of dilute hydrochloric acid on marble chips:



(ii) Industrially it is produced as a by-product during the manufacture of quicklime and in fermentation processes:

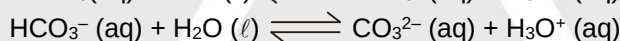
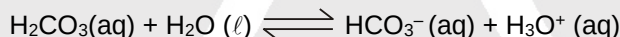


Properties :

(i) It is a colourless, odourless and heavy gas which dissolves in its own volume of water at ordinary temperature and pressure. Like all gases, it dissolves much more readily in water when the pressure is increased and this principle is used in the manufacture of soda water and fizzy drinks.

(ii) CO_2 is easily liquefied (critical temperature = 31.1°C) and a cylinder of the gas under pressure is a convenient fire extinguisher. When the highly compressed gas is allowed to expand rapidly solid carbon dioxide ('dry ice') is formed. Solid carbon dioxide sublimates at -78°C and, since no massy liquid is produced, it is a convenient means of producing low temperatures.

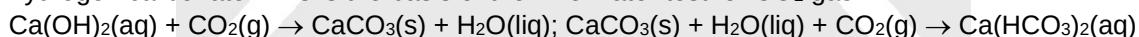
(iii) Carbon dioxide is the acid anhydride of carbonic acid, which is a weak dibasic acid and ionises in two steps as follows :



$\text{H}_2\text{CO}_3 / \text{HCO}_3^-$ buffer system helps to maintain pH of blood between 7.26 to 7.42.

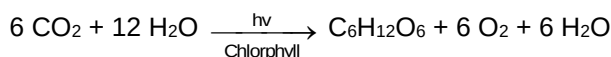
A solution of carbonic acid in water will slowly turn blue litmus red and when the solution is boiled, all the CO_2 is evolved.

(iv) Carbon dioxide readily reacts with alkalis forming the carbonate and, if CO_2 is in excess, the hydrogen carbonate. This is the basis of the lime-water test for CO_2 gas.



The above reaction accounts for the formation of temporarily hard water.

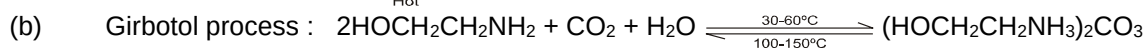
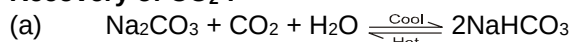
(v) Carbon dioxide, which is normally present to the extent of $\sim 0.03\%$ by volume in the atmosphere, is removed from it by the process known as photosynthesis. It is the process by which green plants convert atmospheric CO_2 into carbohydrates such as glucose. The overall chemical change can be expressed as:



By this process plants make food for themselves as well as for animals and human beings. But the increase in combustion of fossil fuels and decomposition of limestone for cement manufacture in recent years seem to increase the CO_2 content of the atmosphere. This may lead to increase in green house effect and thus, raise the temperature of the atmosphere which might have serious consequences.

(vi) Gaseous CO_2 is extensively used to carbonate soft drinks. Being heavy and non-supporter of combustion it is used as fire extinguisher. A substantial amount of CO_2 is used to manufacture urea.

Recovery of CO_2 :





Solved Examples

Ex-5. $\text{H}_2\text{C}_2\text{O}_4 \xrightarrow{\Delta} \text{gas (A)} + \text{gas (B)} + \text{liquid (C)}$

oxalic acid

Gas (A) burns with blue flame and is oxidised to gas (B). Gas (B) turns lime water milky.

$\text{Gas (A)} + \text{Cl}_2 \longrightarrow (\text{D}) \xrightarrow{\text{NH}_3, \Delta} (\text{E}) \xleftarrow{\text{NH}_3, \Delta} (\text{B})$

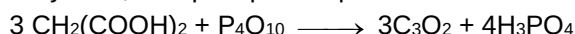
Identify (A) to (E) and explain reactions involved.

Sol. $\text{H}_2\text{C}_2\text{O}_4 \xrightarrow{\Delta} \text{CO} + \text{CO}_2 + \text{H}_2\text{O}$

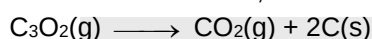
$\text{CO} + \text{Cl}_2 \longrightarrow \text{COCl}_2 \xrightarrow{\text{NH}_3, \Delta} \text{NH}_2\text{CONH}_2 \xleftarrow{\text{NH}_3, \Delta} \text{CO}_2$

Carbon suboxide (C_3O_2) :

This is an evil-smelling gas and can be made by dehydrating propanedioic acid (malonic acid), of which it is the anhydride, with phosphorus pentoxide :



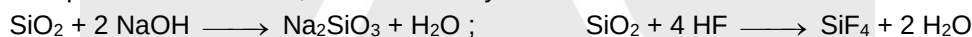
When heated to about 200°C , it decomposes into CO_2 and C:



The molecule is thought to have a linear structure: $\text{O}=\text{C}=\text{C}=\text{O}$.

Silicon Dioxide (SiO_2) :

Silicon dioxide, commonly known as silica, occurs in several crystallographic forms. Quartz, cristobalite and tridymite are some of the crystalline forms of silica, and they are interconvertible at suitable temperature. Silicon dioxide is a covalent, three-dimensional network solid in which each silicon atom is covalently bonded in a tetrahedral manner to four oxygen atoms. Each oxygen atom in turn covalently bonded to another silicon atoms. Each corner is shared with another tetrahedron. The entire crystal may be considered as giant molecule in which eight membered rings are formed with alternates silicon and oxygen atoms. Silica in its normal form is almost non-reactive because of very high Si-O bond enthalpy. It resists the attack by halogens, dihydrogen and most of the acids and metals even at elevated temperatures. However, it is attacked by HF and NaOH.

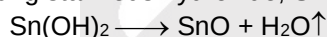


Quartz is extensively used as a piezoelectric material; it has made possible to develop extremely accurate clocks, modern radio and television broadcasting and mobile radio communications. Silica gel used as a drying agent and as a support for chromatographic materials and catalysts. Kieselghur, an amorphous form of silica is used in filtration plants.

Stannous Oxide (SnO) :

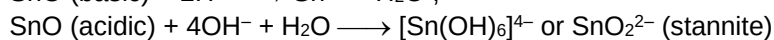
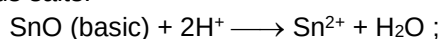
Preparation :

By heating stannous hydroxide, $\text{Sn}(\text{OH})_2$, in absence of air.

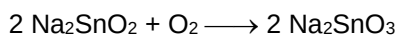


Properties :

SnO is an amphoteric dark grey or black solid oxide, insoluble in water. It dissolves in acids to form stannous salts.



Stannites are only known in aqueous solutions. Stannites absorb oxygen from air and are oxidised to stannate which are stable in nature.



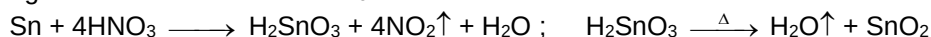
Uses :

For the preparation of stannous chloride and stannous sulphate.

Stannic Oxide (SnO_2) :

Preparation :

By heating tin with concentrated HNO_3 .



**Properties :**

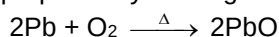
- (i) It is a white solid insoluble in water.
- (ii) It is amphoteric in nature.
- (iii) It dissolves in concentrated H_2SO_4 to form stannic sulphate.

$$\text{SnO}_2 + 2\text{H}^+ \longrightarrow \text{Sn}^{+4} + 2\text{H}_2\text{O}$$
- (iv) It also dissolves in concentrated alkalis to form alkali metal stannate solution.

$$\text{SnO}_2 + 6\text{OH}^- \longrightarrow [\text{Sn}(\text{OH})_6]^{2-} \text{ or } \text{SnO}_3^{2-} \text{ (stannate)}$$

Litharge (PbO) :

PbO is prepared by heating Pb at 180°C . It is a volatile yellow organic solid.



It is an amphoteric oxide and dissolves in acids as well as in alkalis.

It is used in rubber industry and in the manufacture of flint glasses, enamels, and storage batteries.

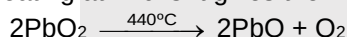
Lead Dioxide (PbO_2) :**Preparation :**

- (i) $\text{PbO} + \text{NaOCl} \xrightarrow{\Delta} \text{PbO}_2 \text{ (insoluble)} + \text{NaCl}$
- (ii) $\text{Pb}_3\text{O}_4 + 4\text{HNO}_3 \text{ (dilute)} \longrightarrow 2\text{Pb}(\text{NO}_3)_2 + \text{PbO}_2 + 2\text{H}_2\text{O}$

Properties :

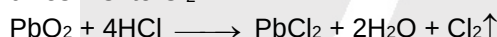
It is a chocolate/dark brown coloured insoluble solid.

- (i) On heating at 440°C it gives the monoxide.



- (ii) PbO_2 is an oxidising agent and reduced to PbO since stability of $\text{Pb(II)} > \text{Pb(IV)}$ based on inert pair effect.

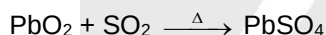
- (a) It oxidizes HCl to Cl_2 .



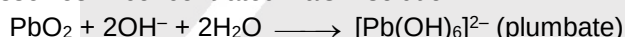
- (b) It oxidises Mn salt to permanganic acid.



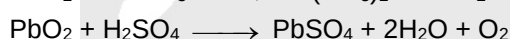
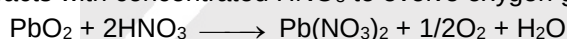
- (c) It reacts with SO_2 at red heat to form lead sulphate.



- (iii) It dissolves in concentrated NaOH solution.



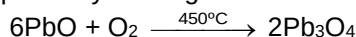
- (iv) It reacts with concentrated HNO_3 to evolve oxygen gas.

**Uses :**

It is used in match industry for making ignition surface of match boxes, in the preparation of KMnO_4 and in explosives.

Red Lead (Pb_3O_4) :**Preparation :**

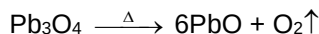
It is prepared by heating PbO at 450°C for a long time.

**Properties :**

- (i) It is a red powder insoluble in water but when heated with concentrated HNO_3 it gives a red precipitate of PbO_2 .

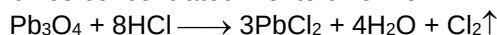


- (ii) When heated above 550°C , it decomposes into PbO.

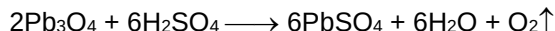




(iii) It oxidizes concentrated HCl to chlorine.



(iv) When heated with concentrated H_2SO_4 it evolves oxygen.



Uses :

It is used as an oxidizing agent, for making metal protecting paints like red oxide paint, for making special lead cement and for making flint glass.

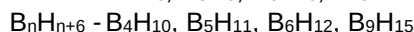
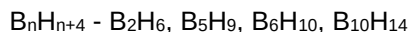
Section (F) : Hydrides

Compounds

Hydrides

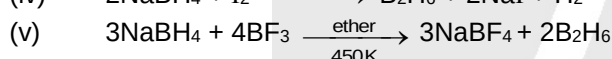
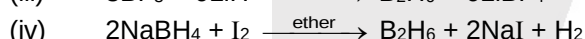
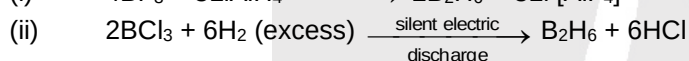
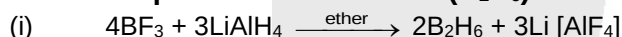
Boranes

Binary compounds of B with H are called boron hydrides or boranes. These compounds form following two types of series :

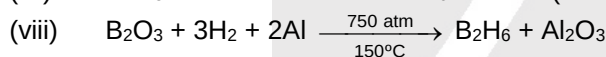
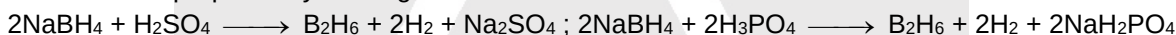


The chemistry of diborane has aroused considerable interest because of its usefulness in many synthetic reactions and also because the elucidation of its structure helped to clarify the basic concepts about the structure of electron deficient compounds.

Preparation of Diborane (B_2H_6) :



(vi) It can also be prepared by treating NaBH_4 with concentrated H_2SO_4 or H_3PO_4 .

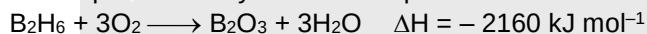


Properties :

(i) B_2H_6 is colourless gas and highly reactive (boiling point 183 K).

(ii) Controlled pyrolysis of diborane leads to most of the higher boranes.

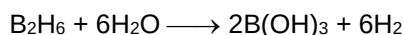
It catches fire spontaneously in air and explodes with O_2 . Reaction with oxygen is extremely exothermic.



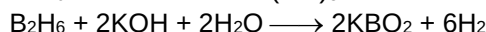
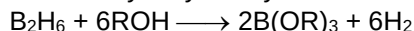
○ Mixtures of diborane with air or oxygen inflame spontaneously producing large amount of heat. Diborane has a higher heat of combustion per unit weight of fuel than most other fuels. It is therefore used as a rocket fuel.

○ At red-heat the boranes decomposes to boron and hydrogen.

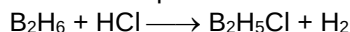
(iii) Reaction with water is instantaneous.



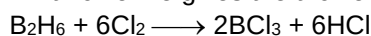
Diborane is also hydrolysed by weaker acids (e.g. alcohols) or aqueous alkali.



(iv) Reaction with HCl replaces a terminal H with Cl.



(v) Reaction with chlorine gives the trichloride.

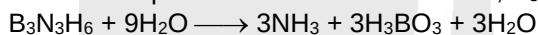


(vi) The electron deficient 3c-2e BHB bridges are sites of nucleophilic attack.

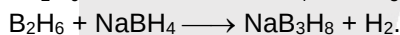
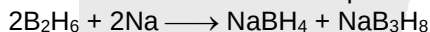


- Small amines such as NH_3 , CH_3NH_2 and $(\text{CH}_3)_2\text{NH}$ give unsymmetrical cleavage of diborane.
 $\text{B}_2\text{H}_6 + 2\text{NH}_3 \longrightarrow [\text{H}_2\text{B}(\text{NH}_3)_2]^+ + [\text{BH}_4]^-$
- Large amines such as $(\text{CH}_3)_3\text{N}$ and pyridine give symmetrical cleavage of diborane.
 $2(\text{CH}_3)_3\text{N} + \text{B}_2\text{H}_6 \longrightarrow 2\text{H}_3\text{B} \longleftarrow \text{N}(\text{CH}_3)_3$
 $\text{B}_2\text{H}_6 + 2\text{Me}_3\text{P} \longrightarrow 2\text{Me}_3\text{PBH}_3$
- $\text{B}_2\text{H}_6 + 2\text{CO} \xrightarrow[20 \text{ atm}]{200^\circ\text{C}} 2\text{BH}_3\text{CO}$ (borane carbonyl)
- The boronium ion products $[\text{H}_2\text{BL}_2]^+$, are tetrahedral and can undergo substitution by other bases
 $[\text{H}_2\text{B}(\text{NH}_3)_2]^+ + 2\text{PR}_3 \longrightarrow [\text{H}_2\text{B}(\text{PR}_3)_2]^+ + 2\text{NH}_3$
- The reaction with ammonia depends on conditions.
 $\text{B}_2\text{H}_6 + \text{NH}_3 \xrightarrow[\text{low temperature}]{\text{Excess NH}_3} \text{B}_2\text{H}_6 \cdot 2\text{NH}_3 \text{ or } [\text{H}_2\text{B}(\text{NH}_3)_2]^+ [\text{BH}_4]^- \text{ (ionic compound).}$
 $\xrightarrow[\text{higher temperature } (> 200^\circ\text{C})]{\text{Excess NH}_3} (\text{BN})_x \text{ boron nitride.}$
 $\xrightarrow[\text{higher temperature } (200^\circ\text{C})]{\text{Ratio } 2\text{NH}_3 : 1 \text{ B}_2\text{H}_6} \text{B}_3\text{N}_3\text{H}_6 \text{ borazine.}$

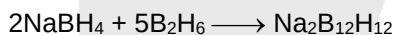
Borazine is much more reactive than benzene. Borazine readily undergoes addition reactions which do not occur with benzene. Borazine also decomposes slowly and may be hydrolysed to NH_3 and boric acid at elevated temperature. If heated with water, $\text{B}_3\text{N}_3\text{H}_6$ hydrolyses slowly.



- (vii) Reduction of diborane can be accomplished with sodium or with sodium borohydride.



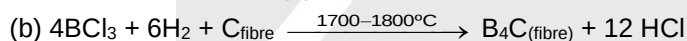
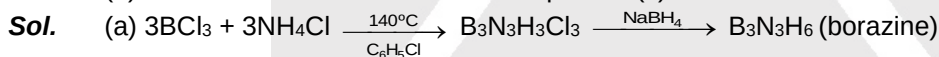
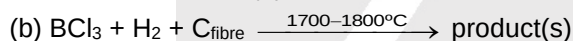
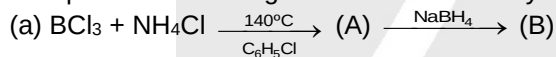
- Reductions of diborane with NaBH_4 can also lead to higher borane anions.



- (viii) $\text{B}_2\text{H}_6 + 2\text{LiH} \longrightarrow 2\text{LiBH}_4$

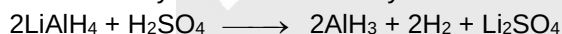
Solved Examples

Ex-6. Complete the following reactions and identify the products formed.



Aluminium Hydride (AlH_3) :

Aluminium hydride is obtained by interaction of LiAlH_4 with 100% H_2SO_4 in THF :



The white hydride is thermally unstable. With donor ligands however, a range of molecular complexes AlH_3L & AlH_3L_2 are formed indicative of the lewis acidic behaviour of AlH_3 .

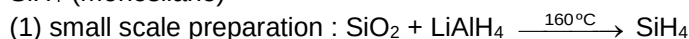
Hydrides of carbon :

Carbon forms a vast number of chain and ring compounds including :

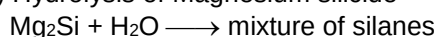
- The alkanes (Paraffins) $\text{C}_n\text{H}_{2n+2}$
- The alkenes (olefines) C_nH_{2n}
- The alkynes (acetylenes) $\text{C}_n\text{H}_{2n-2}$
- Aromatic compounds

Silanes

SiH_4 (monosilane)



(2) Hydrolysis of Magnesium silicide



(3) Reduction of chlorosilanes by LiAlH_4 to produce silane

(4) Photolysis of $\text{SiH}_4\text{--H}_2$ mixture can make higher silanes

(5) Among silanes only SiH_4 & Si_2H_6 are indefinitely stable at 25°C .



Section (G) : Halides

Halides, Alums and Other metal salts :

Boron Halide :

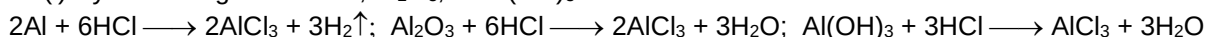
Boron trihalides are strictly monomeric, trigonal planar molecules. This difference from aluminium is attributed to back-donation from “nonbonding” electrons on halogen atoms into the “empty” p_z orbital of boron which lends some double bond character to B–X bonding (manifested by bond shortening) and stabilizes the monomer.

Aluminium Chloride ($\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$) :

It is a colourless crystalline solid, soluble in water. It is covalent. Anhydrous AlCl_3 is a deliquescent white solid.

Preparation :

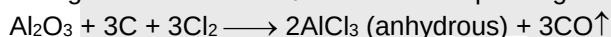
(i) By dissolving aluminium, Al_2O_3 , or $\text{Al}(\text{OH})_3$ in dilute HCl :



The solution obtained is filtered and crystallized when the crystals of $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ are obtained.

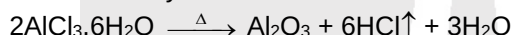
(ii) Anhydrous AlCl_3 is obtained by the action of Cl_2 on heated aluminium.

(iii) By heating a mixture of Al_2O_3 and coke and passing chlorine over it.

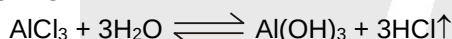


Properties :

(i) **Action of heat** : Hydrated salt when heated strongly is converted to Al_2O_3 .



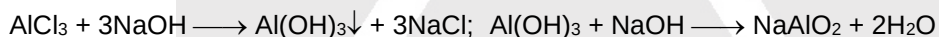
(ii) **Action of moisture on anhydrous AlCl_3** : When exposed to air, anhydrous AlCl_3 produces white fumes of HCl.



(iii) **Action of NH_3** : Anhydrous AlCl_3 absorbs NH_3 since the former is a Lewis acid.

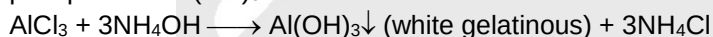


(iv) **Action of NaOH solution** : When NaOH solution is added dropwise to an aqueous AlCl_3 solution, a gelatinous precipitate of $\text{Al}(\text{OH})_3$ is first formed which dissolves in excess of NaOH solution to give a colourless solution of sodium meta-aluminate.



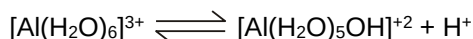
This reaction is important as a test to distinguish between an aluminium salt from salts of Mg, Ca, Sr, and Ba. (When NaOH solution is added to their salt solutions, a white precipitate of hydroxide forms which does not dissolve in excess of NaOH).

(v) **Action of NH_4OH solution** : When NH_4OH solution is added to a solution of AlCl_3 , a white precipitate of $\text{Al}(\text{OH})_3$ is formed which does not dissolve in excess of NH_4OH .

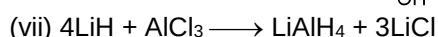
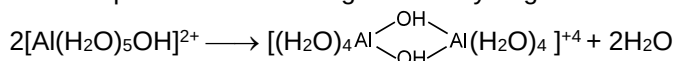


This reaction is important as a test to distinguish an Al salt from a Zn salt. (With a Zn salt a white precipitate of $\text{Zn}(\text{OH})_2$ is formed which dissolves in excess of NH_4OH solution).

(vi) **Hydrolysis with water** : When AlCl_3 is dissolved in water, it undergoes hydrolysis rapidly to produce $\text{Al}(\text{OH})_3$ which is a weak base and HCl which is a strong acid. Hence the solution is acidic to litmus.



The complex cation has a high tendency to get dimerised.



Uses :

- As catalyst for cracking of petroleum.
- As catalyst in Friedel-Crafts reactions.
- For preparing aluminium compounds.



Solved Examples

Ex-7. What happens when :

(write chemical equations only)

(a) Anhydrous AlCl_3 is exposed to atmosphere.

(b) AlCl_3 dissolves in aqueous sodium hydroxide (excess)

Sol. (a) $\text{AlCl}_3 + 3\text{H}_2\text{O} \longrightarrow \text{Al}(\text{OH})_3 + 3\text{HCl}$ (HCl fumes in air).

(b) $\text{AlCl}_3 + 3\text{NaOH} \longrightarrow \text{Al}(\text{OH})_3 \downarrow + 3\text{NaCl}$

$\text{Al}(\text{OH})_3 + \text{OH}^- \longrightarrow [\text{Al}(\text{OH})_4]^-$

Ex-8. Anhydrous AlCl_3 cannot be prepared by heating hydrated salt. Why ?

Sol. Gets hydrolysed forming Al_2O_3 .

$2\text{AlCl}_3 \cdot 6\text{H}_2\text{O} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 6\text{HCl} \uparrow + 3\text{H}_2\text{O}$

Alums ; $\text{M}_2\text{SO}_4 \cdot \text{M}'_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ or $\text{MM}'(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$

Alums are transparent crystalline solids having the above general formula where M is almost any univalent positive cation (except Li^+ because this ion is too small to meet the structural requirements of the crystal) and M' is a trivalent positive cation (Al^{3+} , Ti^{3+} , V^{3+} , Cr^{3+} , Fe^{3+} , Mn^{3+} , Co^{3+} , Ga^{3+} etc.). Alums contain the ions $[\text{M}(\text{H}_2\text{O})_6]^+$, $[\text{M}'(\text{H}_2\text{O})_6]^{3+}$ and SO_4^{2-} in the ratio 1 : 1 : 2. Some important alums are :

(i) Potash alum $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (ii) Chrome alum $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

(iii) Ferric alum $\text{K}_2\text{SO}_4 \cdot \text{Fe}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (iv) Ammonium alum $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

Alums are double salts which when dissolved in water produce metal ions (or ammonium ions) and the sulphate ions.

Preparation :

A mixture containing solutions of M_2SO_4 and $\text{M}'_2(\text{SO}_4)_3$ in 1 : 1 molar ratio is fused & then the resulting mass is dissolved into water. From the solution thus obtained, alums are crystallised.

Uses :

(i) As a mordant in dye industry. The fabric which is to be dyed is dipped in a solution of the alum and heated with steam. $\text{Al}(\text{OH})_3$ obtained as hydrolysis product of $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ deposits into the fibres and then the dye is absorbed on $\text{Al}(\text{OH})_3$.

(ii) As a germicide for water purification

(iii) As a coagulating agent for precipitating colloidal impurities from water.

Solved Examples

Ex-9. List the cations which are capable of replacing aluminium in alums ?

Sol. Cations of about the same size as that of Al^{3+} such as Ti^{3+} , Cr^{3+} , Mn^{3+} , Fe^{3+} and Co^{3+} are capable of replacing aluminium in alums.

Carbon Halides :

CF_4 (Carbon tetrafluoride)

- It is an extraordinarily stable compound.

- Lab preparation : $\text{SiC} + \text{F}_2 \xrightarrow{\Delta} \text{SiF}_4 + \text{CF}_4$

- No hydrolysis is possible.

CCl_4 (Carbon tetrachloride)

- Common solvent, fairly readily decomposed photo chemically.

- Although it is thermodynamically unstable with respect to hydrolysis, the observe of acceptor orbitals carbon makes the attack very difficult.

CBr_4 (Carbon tetrabromide)

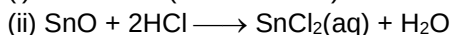
- Pale yellow solid at room temperature.

- Insoluble in water, soluble in non-polar solvent.

CI_4 (Carbon tetraiodide)

- Bright red, crystalline material.

- odor like that of iodine.

**Stannous Chloride (SnCl₂·2H₂O) :****Preparation :**

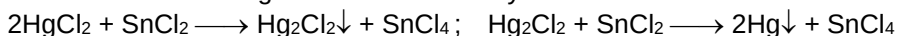
The solution on crystallization gives colourless crystals of SnCl₂·2H₂O.

Properties :

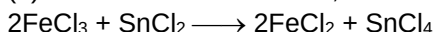
(i) It is a colourless solid soluble in water. It is soluble in alcohol and ether also.

(ii) It is a stronger reducing agent.

(a) Reaction with Hg₂Cl₂ solution: When SnCl₂ solution is added to an aqueous solution of mercuric chloride, a silky white precipitate of mercurous chloride, Hg₂Cl₂ is formed which turns black due to further reduction of Hg₂Cl₂ to black mercury.

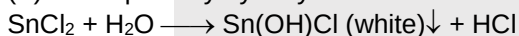


(b) It reduces ferric chloride, FeCl₃ to ferrous chloride, FeCl₂.



(c) It also reduces CuCl₂ to CuCl (white).

(iii) SnCl₂ partially hydrolyses in water forming the basic chloride, Sn(OH)Cl.



As it produces a weak base and strong acid its aqueous solution is acidic. Its hydrolysis can be prevented by adding concentrated HCl to it during the process of its preparation.

Uses :

(i) In dye industry as a reducing agent.

(ii) For the test of mercuric salt.

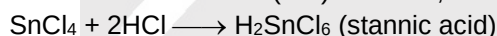
(iii) For the preparation of other stannous compounds.

Stannic Chloride (SnCl₄) :**Preparation :****Properties :**

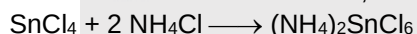
(i) It is a colourless fuming liquid ; boiling point is 114°C.

(ii) Action of moisture: It absorbs moisture and becomes converted into hydrated stannic chlorides, SnCl₄·3H₂O, SnCl₄·5H₂O, SnCl₄·6H₂O and SnCl₄·8H₂O. SnCl₄·5 H₂O is known as "butter of tin" or "oxymercurate of tin".

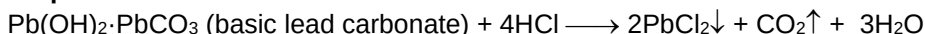
(iii) Hydrolysis with water : It hydrolyses in dilute solution but it is incomplete and can be repressed in presence of halogen acid.



(v) In presence of ammonium chloride, it forms ammonium salt of H₂SnCl₆ (stannic acid).

**Uses :**

For the preparation of stannic compounds.

Lead Chloride (PbCl₂) :**Preparation:****Properties :**

It is a white crystalline solid, insoluble in cold water but soluble in boiling water. It dissolves in concentrated HCl forming a complex ion.

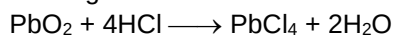
**Uses :**

It is used for making pigments for paints.

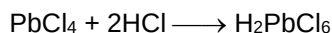
**Lead Tetrachloride (PbCl₄) :****Preparation :**

It is prepared by the following methods:

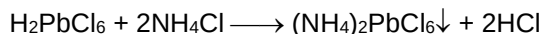
(i) By dissolving PbO₂ in cold concentrated HCl



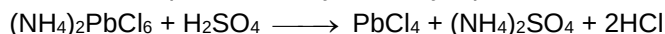
PbCl₄ dissolves in excess of HCl to form a stable solution of H₂PbCl₆.



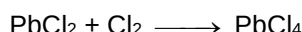
When NH₄Cl is added to a solution of chloroplumbic acid, a yellow precipitate of ammonium chloroplumbate is formed.



When crystals of ammonium chloroplumbate is added to ice cold concentrated H₂SO₄, lead tetrachloride is formed and separates as a yellow oily liquid.



(ii) By the action of Cl₂ on a solution of PbCl₂ in concentrated HCl

**Properties :**

(i) It is a yellow oily liquid which solidifies at -10°C and is soluble in organic solvents like ethanol and benzene.

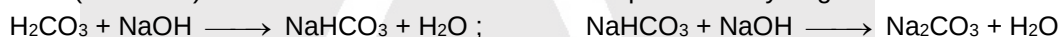
(ii) Rapid hydrolysis with water forms PbO₂ precipitate

**Uses :**

It is used for making stannic compounds.

Carbonates (CO₃²⁻) and Bicarbonates (HCO₃⁻)

Carbonic acid is a dibasic acids giving rise to two series of salts, carbonates (normal salts) and bicarbonates (acid salts) due to successive removal of the replaceable hydrogens from H₂CO₃.

**Preparation :**

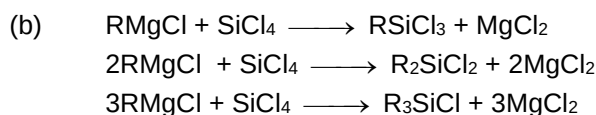
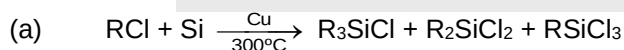
(i) With NaOH : $2\text{NaOH} + \text{CO}_2 \longrightarrow \text{Na}_2\text{CO}_3$; $\text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2 \longrightarrow 2\text{NaHCO}_3$

(ii) By precipitation : $\text{BaCl}_2 + \text{Na}_2\text{CO}_3 \longrightarrow \text{BaCO}_3 \downarrow + 2\text{NaCl}$

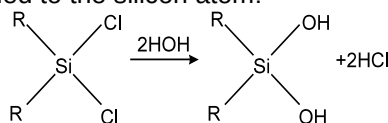
SiCl₄, Silicones, Silicates & Zeolites :**Silicones :**

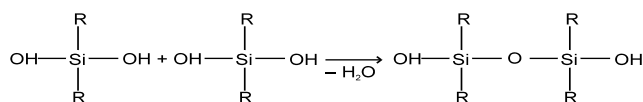
Silicones are synthetic organosilicon compounds having repeated R₂SiO units held by Si-O-Si linkages. These compounds have the general formula (R₂SiO)_n where R = alkyl or aryl group.

The silicones are formed by the hydrolysis of alkyl or aryl substituted chlorosilanes and their subsequent polymerisation. The alkyl or aryl substituted chlorosilanes are prepared by the following reactions.

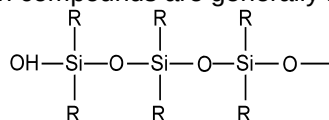


After fractional distillation, the silane derivatives are hydrolysed and the 'hydroxides' immediately condense by intermolecular elimination of water. The final product depends upon the number of hydroxyl groups originally bonded to the silicon atom:





In this manner several molecules may combine to form a long chain polymer whose both the ends will be occupied by $-\text{OH}$ groups. Such compounds are generally represented from the following formula.

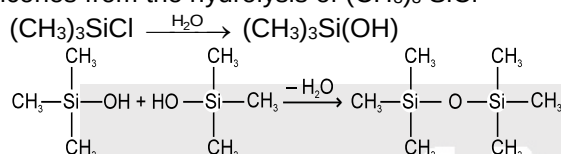


The polymer chain depicted above is terminated by incorporating a small quantity of the monochlorosilane derivative into the hydrolysis mixture.

- Silicones can be prepared from the following types of compounds only.

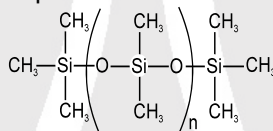
(i) R_3SiCl (ii) R_2SiCl_2 (iii) RSiCl_3

- Silicones from the hydrolysis of $(\text{CH}_3)_3\text{SiCl}$



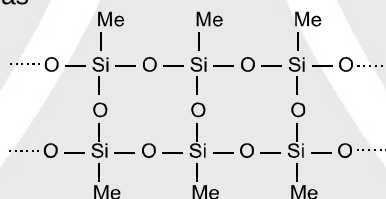
- Silicones from the hydrolysis of a mixture of $(\text{CH}_3)_3\text{SiCl}$ & $(\text{CH}_3)_2\text{SiCl}_2$

The dichloro derivative will form a long chain polymer as usual. But the growth of this polymer can be blocked at any stage by the hydrolysis product of mono-chloro derivative.



- Silicones from the hydrolysis of trichloro derivative.

When a compound like CH_3SiCl_3 undergoes hydrolysis, a complex cross-linked polymer is obtained as chain can grow in three places as



- The hydrocarbon layer along the silicon-oxygen chain makes silicones water-repellent.
- Silicones find a variety of applications because of their chemical inertness, water repelling nature, heat resistance and good electrical insulation property.

Products having the physical properties of oils, rubbers and resins can be produced using silicones. Silicone varnishes are such excellent insulators and so heat-resistance that insulating wiring with them enabled motors to work over-loads that would have set fire to the insulation formerly used. Silicone fluids are used as hydraulic systems of planes as they are thermally stable and their viscosity alters very little with temperature. Silicone rubbers are used in place of ordinary rubber as they retain their elasticity at much lower temperature than ordinary rubber.

Solved Examples

Ex-10. Complete the following reactions

- (a) $\text{CO} + \text{H}_2 \xrightarrow[420-670\text{k}, 300\text{atm}]{\text{ZnO}+\text{Cu}} \dots\dots\dots$
- (b) $\text{R}_3\text{SiOH} + \text{OHSiR}_3 \longrightarrow \dots\dots\dots + \dots\dots\dots$
- (c) $\text{Na}_2\text{CO}_3 + \text{Si} \longrightarrow \dots\dots\dots + \dots\dots\dots$

Sol. (a) $\text{CO} + 2\text{H}_2 \xrightarrow[420-670\text{k}, 300\text{atm}]{\text{ZnO}+\text{Cu}} \text{CH}_3\text{OH}$

(b) $\text{R}_3\text{SiOH} + \text{OHSiR}_3 \longrightarrow \text{R}_3\text{Si}-\text{O}-\text{SiR}_3 + \text{H}_2\text{O}$

(c) $\text{Na}_2\text{CO}_3 + \text{Si} \longrightarrow \text{Na}_2\text{SiO}_3 + \text{C}$





Section (H) : Miscellaneous (Silicones, Silicates, Zeolites & Alums)

Silicates :

Binary compounds of silicon with oxygen are called silicates but they contain other metals also in their structures.

- (i) Since the electronegativity difference between O & Si is about 1.7, so Si–O bond can be considered 50% ionic & 50% covalent.

- (ii) If we calculate the radius ratio then, $\frac{r_{\text{Si}^{4+}}}{r_{\text{O}^{2-}}} = 0.29$

It suggests that the coordination number of silicon must be 4 and from VBT point of view we can say that Si is sp^3 hybridized. Therefore silicate structures must be based upon SiO_4^{4-} tetrahedral units.

- (iii) SiO_4^{4-} tetrahedral units may exist as discrete units or may polymerise into larger units by sharing corners.

Classification of Silicates :

(A) Orthosilicates :

These contain discrete $[\text{SiO}_4]^{4-}$ units i.e., there is no sharing of corners with one another as shown in figure.

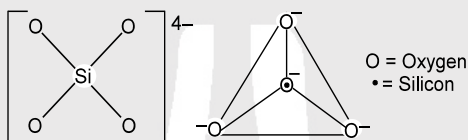


Figure : 2

e.g. Zircon (ZrSiO_4), Forsterite or Olivine (Mg_2SiO_4), Willemite (Zn_2SiO_4)

(B) Pyrosilicate :

In these silicates two tetrahedral units are joined by sharing oxygen at one corner thereby giving $[\text{Si}_2\text{O}_7]^{6-}$ units.

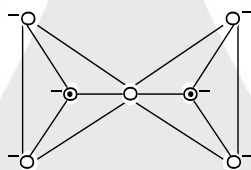


Figure : 3

e.g. Thortveitite ($\text{Sc}_2\text{Si}_2\text{O}_7$), Hemimorphite ($\text{Zn}_3(\text{Si}_2\text{O}_7) \cdot \text{Zn}(\text{OH})_2 \cdot \text{H}_2\text{O}$)

○ (–) charge will be present on the oxygen atoms which is bonded with one Si atom.

(C) Cyclic silicates :

If two oxygen atoms per tetrahedron are shared to form closed rings such that the structure with general formula $(\text{SiO}_3^{2-})_n$ or $(\text{SiO}_3)_n^{2n-}$ is obtained, the silicates containing these anions are called cyclic silicates. $\text{Si}_3\text{O}_9^{6-}$ and $\text{Si}_6\text{O}_{18}^{12-}$ anions are the typical examples of cyclic silicates.

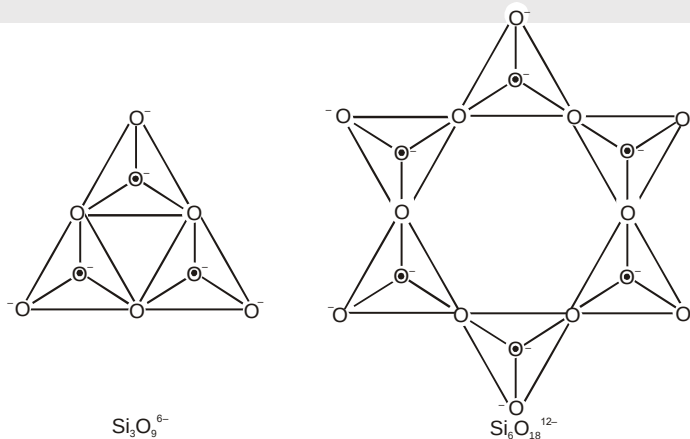


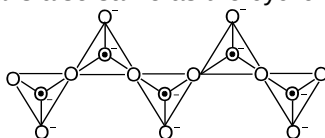
Figure : 4

Figure : 5

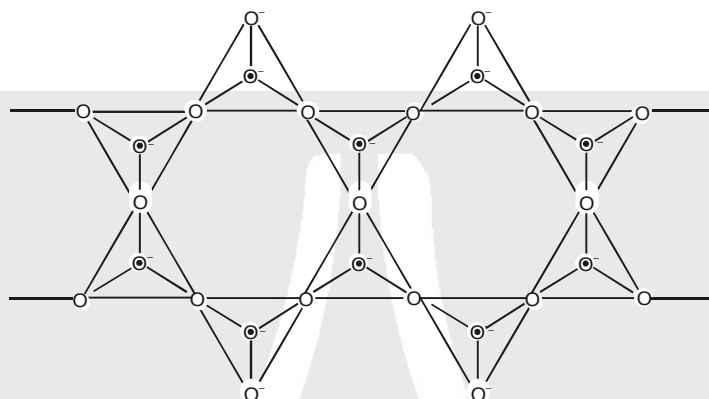
**(D) Chain silicates :**

Chain silicates may be further classified into simple chain & double chain compounds.

In case of simple chains two corners of each tetrahedron are shared & they form a long chain of tetrahedron. Their general formula is also same as the cyclic silicates i.e. $(\text{SiO}_3)_n^{2n-}$

**Figure : 6**

Similarly, double chain silicates can be drawn in which two simple chains are joined together by shared oxygen. Such compounds are also known as amphiboles. The asbestos mineral is a well known example of double chain silicates. The anions of double chain silicates have general formula $(\text{Si}_4\text{O}_{11})_n^{6n-}$

**Figure : 7**

e.g., Synthetic silicates (Li_2SiO_3 , Na_2SiO_3), Spondumene ($\text{LiAl}(\text{SiO}_3)_2$), Enstatite (MgSiO_3), Diopside ($\text{CaMg}(\text{SiO}_3)_2$), Tremolite ($\text{Ca}_2\text{Mg}_5(\text{Si}_4\text{O}_{11})_2(\text{OH})_2$), etc.

(E) Two dimensional sheet silicates :

In such silicates, three oxygen atoms of each tetrahedral are shared with adjacent SiO_4^{4-} tetrahedrals. Such sharing forms two dimension sheet structure with general formula $(\text{Si}_2\text{O}_5)_n^{2n-}$
e.g. Talc ($\text{Mg}(\text{Si}_2\text{O}_5)_2 \text{Mg}(\text{OH})_2$), Kaolin $\text{Al}_2(\text{OH})_4(\text{Si}_2\text{O}_5)$

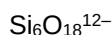
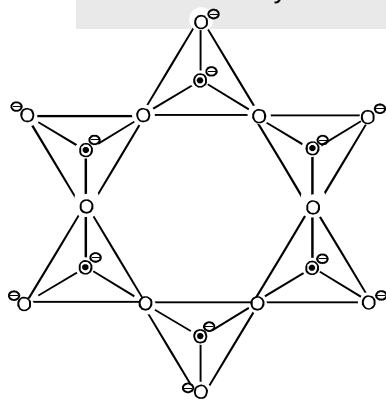
(F) Three dimensional silicates :

These silicates involve all four oxygen atom in sharing with adjacent SiO_4^{4-} tetrahedral units.
e.g. Quartz, Tridymite, Cristobalite, Feldspar, Zeolite and Ultramarines.

Solved Examples

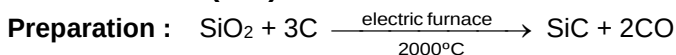
Ex-11. Draw the structure of cyclic silicate containing $\text{Si}_6\text{O}_{18}^{12-}$ ion.

Sol.

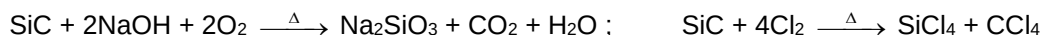


**Zeolites :**

If aluminium atoms replace few silicon atoms in three-dimensional network of silicon dioxide, overall structure known as aluminosilicate, acquires a negative charge. Cations such as Na^+ , K^+ or Ca^{2+} balance the negative charge. Examples are feldspar and zeolites. Zeolites are widely used as a catalyst in petrochemical industries for cracking of hydrocarbons and isomerisation, e.g., ZSM-5 (A type of zeolite) used to convert alcohols directly into gasoline. Hydrated zeolites are used as ion exchangers in softening of "hard" water.

Carborundum (SiC) :**Properties :**

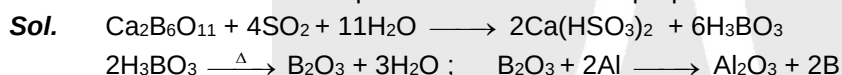
- (i) It is a very hard substance (Hardness = 9.5 Moh)
- (ii) On heating it does not melt rather decomposes into elements.
- (iii) Not attacked by acids. However, it gives the following two reactions at high temperature.



- It has a diamond like structure in which each atom is sp^3 hybridized. Therefore, each atom is tetrahedrally surrounded by 4 atoms of other type.

Solved Examples

Ex-12. Write the chemical equations involved in the preparation of elemental boron from mineral colemanite.

**Uses of boron :**

- (i) Boron is used in the construction of high impact-resistant steel and, since it absorbs neutrons, in reactor rods for controlling atomic reactions.
- (ii) Boron carbide is used as an abrasive.

Uses of Al :

It is extensively used :

- (i) for manufacture of cooking and household utensils.
- (ii) as aluminium plating for tanks, pipes, iron bars and other steel objects to prevent corrosion.
- (iii) for manufacture of aluminium cables.
- (iv) for making precision instruments, surgical apparatus, aircraft bodies, rail coaches, motorboats, car.
- (v) Aluminates are important constituents of portland cement.

Uses of carbon : Graphite fibres embedded in plastic material form high strength, lightweight composites. The composites are used in products such as tennis rackets, fishing rods, aircraft and canoes. Being good conductor, graphite is used for electrodes in batteries and industrial electrolysis. Crucibles made from graphite are inert to dilute acids and alkalis. Being highly porous and having enormous surface area activated charcoal is used in adsorbing poisonous gases; also used in water filters to remove organic contaminants and in air conditioning system to control odour. Carbon black is used as black pigment in black ink and as filler in automobile tyres. Coke is used as a fuel and largely as a reducing agent in metallurgy. Diamond is a precious stone and used in jewellery. It is measured in carats (1 carat = 200 mg.).

Uses of silicon :

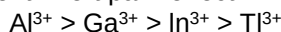
- (i) Ultrapure form of germanium and silicon are used to make transistors and semiconductor devices.
- (ii) Silicon is a very important component of ceramics, glass and cement.



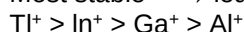
MISCELLANEOUS SOLVED PROBLEMS (MSPS)

1. Al and Ga are trivalent in their compounds but monovalent compounds are the most stable down the 13th group. Why ?

Sol. Down the group (13th), the stability of +3 state decreases and that of +1 state increases due to the prominent "inert pair" effect.

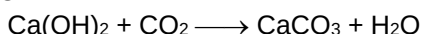


Most stable $\longrightarrow$ least stable

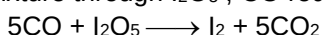


2. If you have a mixture of CO and CO₂, how would you know about the relative proportions of the two gases in the given mixture ?

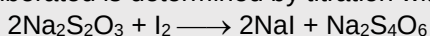
Sol. (i) Pass mixture through the Ca(OH)₂ solution; CO₂ is absorbed by Ca(OH)₂. The residual volume will be that of CO



(ii) Pass mixture through I₂O₅; CO reduces I₂O₅ to I₂.



I₂ thus liberated is determined by titration with Na₂S₂O₃.



This is the quantitative method of estimation of CO.

3. What will happen if borontrifluoride is kept in moist air ?

(A) It will strongly fume.

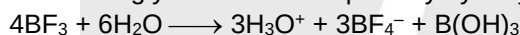
(B) It will partially hydrolyse.

(C) It will completely hydrolyse.

(D) None of these

Ans. (A)

Sol. In moist air it strongly fume : but it is partially hydrolysed by excess of water.



BF₃ is a colourless gas.

4. What happens when : (write only chemical reactions)

(a) iodine is treated with SnCl₂.

(b) carbondioxide is passed through a concentrated aqueous solution of sodium chloride saturated with ammonia.

(c) red lead is treated with nitric acid.

(d) dilute nitric acid is slowly reacted with tin.

Sol. (a) $2\text{SnCl}_2 + \text{I}_2 \longrightarrow 2\text{SnCl}_2\text{I}_2 \longrightarrow \text{SnCl}_4 + \text{SnI}_4$

(b) $\text{NaCl} + \text{NH}_4\text{OH} + \text{CO}_2 \longrightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$

(c) $\text{Pb}_3\text{O}_4 + 4\text{HNO}_3 \longrightarrow 2\text{Pb(NO}_3)_2 + \text{PbO}_2 + 2\text{H}_2\text{O}$

(d) $\text{Sn} + 10\text{HNO}_3(\text{dilute}) \longrightarrow 4\text{Sn(NO}_3)_2 + \text{NH}_4\text{NO}_3 + 3\text{H}_2\text{O}$

5. True / False

(a) BCl₃ in aqueous solution exists as B³⁺ and Cl⁻.

(b) Pure crystalline boron is very unreactive and it is attacked only at high temperatures by strong oxidising agents such as a mixture of hot concentrated H₂SO₄ and HNO₃.

(c) AlX₃ (X = Cl, Br) exists as dimer and retains dimer formula in non-polar solvents like ether, benzene etc.

(d) Be₂C is called acetylide because it reacts with water yielding ethyne.

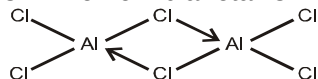
(e) Pb₃O₄ a double oxide, is obtained by heating lead (II) oxide in air.

Ans. (a) False (b) True (c) True (d) False (e) True

Sol. (a) Statement is incorrect. BCl₃ hydrolyses in aqueous solution to give boric acid. Because it has large ionisation energies and to make the enthalpy of solution of BCl₃ negative, the enthalpy of hydration of B³⁺ should be very high (~ 600 kJ) which is unlikely for the small B³⁺ cation.

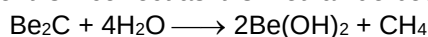
(b) $2\text{B} + 6\text{HNO}_3(\text{aq.}) \longrightarrow 2\text{H}_3\text{BO}_3(\text{aq.}) + 6\text{NO}_2(\text{g})$

(c) Statement is correct and its dimer structure is as follows. It acquires this structure for attaining an octet of electrons. Dimer formula retains in non-polar solvent like ether, benzene





- (d) Statement is incorrect as it is methanide because it gives methane on reaction with water.



- (e) $3\text{PbO} + \text{O}_2 \xrightarrow{\Delta} \text{Pb}_3\text{O}_4$

6. **Statement-1** : PbO_2 is an oxidising agent and reduced to PbO .

Statement-2 : Stability of $\text{Pb(II)} > \text{Pb(IV)}$ on account of inert pair effect.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

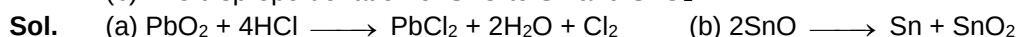
Ans. (A)

Sol. Both are correct statements and statement-2 is the correct explanation of statement-1.

7. Write the chemical equations to represent the following reactions.

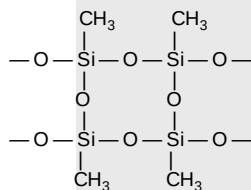
(a) The oxidation of HCl (aq) to Cl_2 (g) by PbO_2 .

(b) The disproportionation of SnO to Sn and SnO_2 .



8. What will happen if we take $\text{Si}(\text{CH}_3)\text{Cl}_3$ as a starting material for the preparation of commercial silicon polymer?

Sol. With $\text{Si}(\text{CH}_3)\text{Cl}_3$ the chain will grow in three places and we will get cross-linked silicon polymer as shown below :



9. Give three properties of diamond.

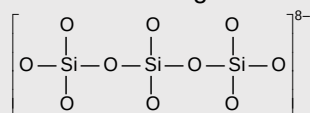
Sol. Diamond is very hard, high melting solid. It is an electrical insulator.

10. The silicate anion in the mineral kionite is a chain of three SiO_4 tetrahedra that share corners with adjacent tetrahedra. The mineral also contains Ca^{2+} ions, Cu^{2+} ions, and water molecules in a 1:1:1 ratio.

(a) Give the formula and charge of the silicate anion.

(b) Given the complete formula for the mineral.

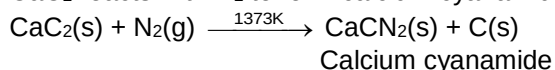
Sol. (a) The silicate anion has three SiO_4 tetrahedra that share corners with adjacent tetrahedra thus silicate is Si_3O_{10} , hence it can be represented as with charge as $= 3 \times 4n + 10 \times (-2) = -8$



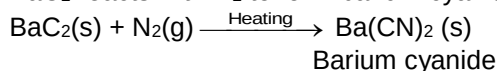
(b) Ca^{2+} , Cu^{2+} and H_2O are in the ratio of 1 : 1 : 1 and to balance (-8) charge of silicate as ion, $(+8)$ charge is required thus there are two units each of Ca^{2+} , Cu^{2+} and H_2O thus, kionoite has formula $\text{Ca}_2\text{Cu}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$.

11. In what respect the reaction of N_2 with (i) CaC_2 (calcium carbide) & (ii) BaC_2 (barium carbide) differ from each other.

Sol. (i) CaC_2 reacts with N_2 to form calcium cyanamide.

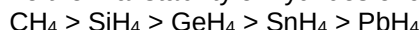


(ii) BaC_2 reacts with N_2 to form barium cyanide





12. **Statement-1** : The thermal stability of hydrides of carbon family is in order :



Statement-2 : E—H bond dissociation enthalpies of the hydrides of carbon family decrease down the group with increasing atomic size.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

Ans. (A)

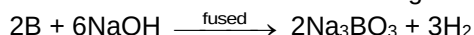
Sol. Both are correct statements and statement-2 is the correct explanation of statement-1. Down the group the size of atom increases and thus bond length increases.

13. Which one of the following element does not dissolve in fused or aqueous alkalis?

- (A) Boron (B) Silicon (C) Aluminium (D) None of these

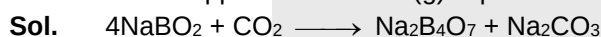
Ans. (D)

Sol. Boron dissolved in fused alkalis according to the following reaction.



Silicon and aluminium dissolved in both fused and aqueous alkalis.

14. What happens when CO_2 (g) is passed through sodium meta borate solution ?

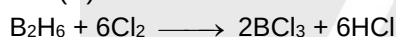


15. Which of the following statement(s) is/are correct ?

- (A) B_2O_3 and SiO_2 are acidic in nature and are important constituents of glass.
 (B) Borides and silicides are hydrolysed by water forming boranes and silanes respectively.
 (C) Diborane on reaction with chlorine (g) forms $\text{B}_2\text{H}_5\text{Cl}$.
 (D) SiO_4^{4-} gets hydrolysed by acid or water and form $\text{Si}_2\text{O}_7^{6-}$.

Ans. (A), (B) and (D)

Sol. (A), (B) and (D) are correct statements but (C) is incorrect.

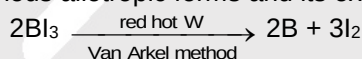


16. Match the following :

	Column-I		Column-II
(A)	Boron	(p)	Forms acidic oxides.
(B)	Carbon	(q)	Pure crystalline form is obtained by Van Arkel method.
(C)	Tin	(r)	Exists in allotropic forms.
(D)	Aluminium	(s)	Hydroxide is amphoteric in nature.

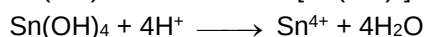
Ans. (A - p,q,r); (B - p,r); (C - r,s); (D - s)

Sol. (A) Exists in various allotropic forms and its oxide, B_2O_3 is acidic in nature.



(B) Exists in various allotropic forms like diamond, graphite etc. and its oxide CO_2 is acidic in nature.

(C) Exists in allotropic forms like grey tin (α -Sn) and white tin (β -Sn). Hydroxide is amphoteric in nature.



(D) Hydroxide is amphoteric in nature.

